

EXP 23:

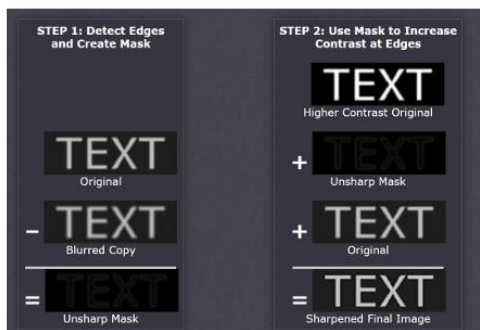
Perform Sharpening of Image using unsharp masking.

Unsharp masking

$$f_s(x, y) = f(x, y) - \bar{f}(x, y)$$

sharpened image = original image – blurred image

- to subtract a blurred version of an image produces sharpening output image.



AIM:

Sharpening of Image using unsharp masking.

Program:

```
import cv2

import numpy as np

img = cv2.imread(r'C:\Users\pavan\OneDrive\Opencv\x.jpg')

if img is None:

    print("Error: Image not found or path is incorrect")

else:

    kernel = np.array([[ -1, -1, -1],
                       [-1,  8, -1],
                       [-1, -1, -1]])

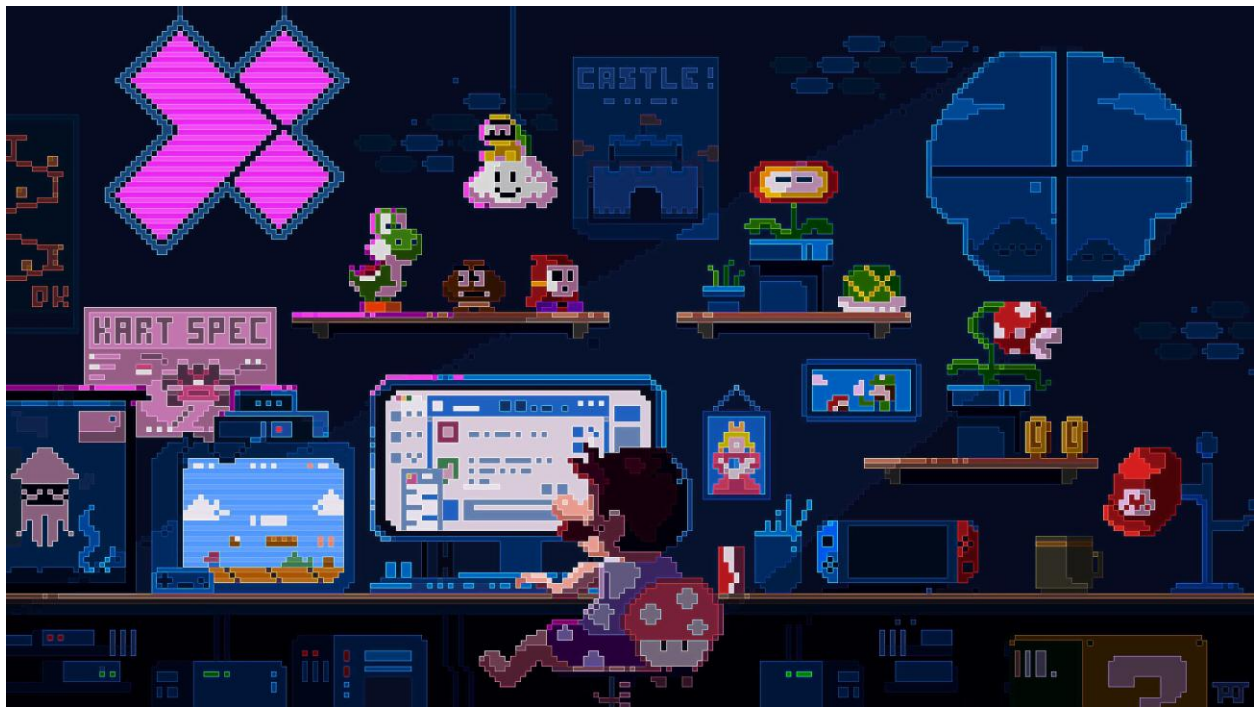
    laplacian = cv2.filter2D(img, -1, kernel)

    sharpened = cv2.add(img, laplacian)
```

INPUT:



Output:



RESULT:- Thus, the program is executed successfully.